

Entropy Regularization, RND, and ICM: A Comparison of Exploration Mechanisms for On and Off-Policy RL Under Sparse Rewards

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1. Motivation

Sparse rewards remain one of the central challenges in deep reinforcement learning: an agent may interact with an environment for thousands of steps before receiving any learning signal, making policy optimization slow and unstable. Three families of solutions have emerged — *entropy regularization*, which encourages exploratory stochastic policies by adding a policy-entropy bonus; *Random Network Distillation (RND)*, which augments the extrinsic reward with a novelty bonus derived from a fixed random network’s prediction error; and the *Intrinsic Curiosity Module (ICM)*, which uses a forward/inverse dynamics model to generate curiosity-driven intrinsic rewards. All three are widely used in practice, yet they are rarely evaluated together under matched compute and controlled reward-density conditions.

A successful project would enable practitioners to make principled choices when deploying on-policy (PPO) or off-policy (DQN) agents in sparse-reward settings, targeting a concrete metric: identifying the regime where RND or ICM reduces episodes-to-threshold by $\geq 20\%$ relative to entropy-only baselines, and whether ICM’s dynamics-based signal offers advantages over RND’s simpler prediction-error approach.

2. Gap in the Literature

Prior work. Mnih et al. (2015) established DQN as a strong off-policy baseline on discrete-action tasks. Schulman et al. (2017) introduced PPO with clipped surrogate objectives. Pathak et al. (2017) proposed ICM, generating curiosity-driven intrinsic rewards via a forward/inverse dynamics model. Burda et al. (2019) proposed RND as a simpler, more scalable alternative based on fixed random network prediction error. These works differ in environments, reward structures, and compute budgets, making direct comparison difficult.

What is missing. No controlled study jointly isolates entropy regularization, RND, and ICM across both on-policy and off-policy algorithms under matched compute and identical reward-density conditions. Prior comparisons either fix the algorithm family or evaluate exploration mechanisms in isolation without a cross-algorithm perspective.

Our angle. *No controlled study compares entropy regularization, RND, and ICM across PPO and DQN under matched wall-clock compute on Gymnax classic-control environments with both sparse and dense reward variants.*

3. Research Questions

Primary: Under sparse rewards and matched compute, do RND or ICM intrinsic motivation methods improve sample efficiency (episodes to threshold return) more for on-policy PPO than for off-policy DQN, or is the gap closed by entropy regularization alone?

Secondary: Does ICM’s dynamics-based curiosity signal yield different sample-efficiency gains compared to RND’s prediction-error bonus, and does this difference interact with the on-policy vs. off-policy distinction as reward density varies?

4. Experimental Methodology

Environment & Data. We use Gymnax (JAX-native) CartPole-v1 and MountainCar-v0. Both support fast, parallelisable rollouts. Sparse-reward variants replace the dense reward with a binary success/failure signal at episode end, controlling reward density as an independent variable.

Algorithms & Conditions (8 total).

- **PPO** — on-policy baseline
- **PPO + Entropy Regularization** ($\alpha \in \{0.01, 0.05\}$)
- **PPO + RND** — prediction-error intrinsic reward
- **PPO + ICM** — curiosity via forward/inverse dynamics model
- **DQN** — off-policy baseline
- **DQN + Entropy Regularization**

- **DQN + RND**
- **DQN + ICM**

All conditions use identical MLP architectures (2 hidden layers, 64 units, ReLU) and the same wall-clock budget (2 hours per condition on a personal GPU).

Controls & Baselines. A random-action agent and a hand-coded heuristic agent serve as lower bounds. Vanilla PPO and DQN are primary baselines. All methods share identical evaluation protocols (greedy rollouts every 50 training episodes).

Hyperparameter Search. Learning rates $\in \{1e-4, 1e-3\}$; discount $\gamma = 0.99$; entropy coefficient $\alpha \in \{0.01, 0.05\}$; RND reward scale $\beta \in \{0.1, 1.0\}$; ICM curiosity scale $\eta \in \{0.1, 1.0\}$. Selection by validation return on a held-out 20% of seeds; fixed budget of 40 configurations.

Seeds & Statistics. 10 independent seeds per condition. Results reported as mean $\pm$ standard error with individual seed traces. Welch's t-test used for pairwise significance.

Metrics. Average return vs. episodes; episodes-to-threshold (90% of max return); policy entropy over training; RND/ICM intrinsic reward magnitude over training.

Compute & Reproducibility. Personal GPU (NVIDIA). JAX + Flax; pinned versions in environment.yml. Public GitHub repo with README, deterministic run scripts, and pre-trained checkpoints. All experiments reproducible from a single shell command.

GitHub Link: https://github.com/mohammad1774/elg5214_group_8_project

Risk & Mitigation. Risk: DQN training instability in sparse MountainCar and ICM overhead increasing wall-clock time. Mitigation: gradient clipping (norm 10), prioritised experience replay, lightweight ICM network (1 hidden layer, 64 units), and fallback to CartPole-only if needed.

5. Expected Contributions

- A reproducible, matched-compute comparison clarifying when RND or ICM outperforms entropy regularization for both PPO and DQN under sparse and dense rewards.
- Clean JAX implementations of PPO, DQN, RND, ICM, and entropy regularization, reusable as community baselines.
- An ablation over reward density and exploration mechanism revealing crossover points where intrinsic motivation ceases to provide a statistically significant advantage.

6. Work Distribution

Student A will lead the literature review and finalise the research questions. **Student B & C** will implement and validate the PPO variants (baseline, entropy, RND, ICM). **Student A & D** will implement and validate the DQN variants (baseline, entropy, RND, ICM). **All members** will contribute to experiment design, hyperparameter sweeps, statistical analysis, and report writing. Code review is cross-assigned (A reviews D's code, B reviews C's code and vis a vis) to ensure correctness.

References

- [1] Mnih, V. et al. (2015). Human-level control through deep reinforcement learning. *Nature*, 518, 529–533.
- [2] Schulman, J. et al. (2017). Proximal Policy Optimization Algorithms. arXiv:1707.06347.
- [3] Pathak, D. et al. (2017). Curiosity-driven Exploration by Self-Supervised Prediction. ICML 2017.
- [4] Burda, Y. et al. (2019). Exploration by Random Network Distillation. ICLR 2019.
- [5] Lax, M. et al. (2023). Gymnax: A JAX-based Reinforcement Learning Environment Library. arXiv:2311.16943.